

# Pulse-Response Characteristics of Position-Sensitive Photodetectors

CLAUDE A. KLEIN, MEMBER, IEEE, AND ROBERT W. BIERIG, SENIOR MEMBER, IEEE

**Abstract**—A key factor in designing position-sensitive detectors operating through lateral photoeffects in semiconductors concerns the nature of the pulse response over the entire active area of the device. It is demonstrated that the limiting behavior can be approximated by means of a fundamental  $RC$ -filter time constant  $\tau_c$  together with position-dependent initial delay times of up to  $0.6\tau_c$ . A comparison with recorded pulse-response traces suggests that the time constant  $\tau_c$  of two-dimensional position-sensitive photodetectors takes on values of  $R_S C_A d^2 / \pi^2$ , where  $R_S$  is the backlayer sheet resistance,  $C_A$  the junction capacitance per unit area, and  $d$  the electrode separation. Illustrations of the dependence of pulse shape on spot position for representative ion-implanted silicon units are shown to substantiate the analysis in the context of evaluating position-sensitive photodetectors for applications involving very short laser pulses.

## I. INTRODUCTION

**P**osition-sensitive photodetectors (PSD's) are semiconductor devices that make use of the lateral photoeffect to deliver four electrical signals per incident light pulse. These signals can be processed to extract both "horizontal" and "vertical" components of the angle of incidence if the associated optical assembly focuses the incoming light to a small spot on the detector face, at a position proportional to the angle of incidence [1]. The lateral photocurrent, which results from the nonuniform illumination of a rectifying junction is then detected at four sensing electrodes suitably located at the edge of the active area; after amplification and ratiometric amplitude processing, these signals provide two dc analog outputs that are linearly related to the spot position.

A key factor in the effective utilization of these devices concerns the nature of the pulse response over the entire face of the detector. Specifically, in considering PSD's for operation in conjunction with very short laser pulses it is desirable that the detected pulse shape, as seen at the terminating impedances, remains independent of the light-impact position with respect to the sensing electrodes. In this connection, it is essential to keep in mind that the frequency response of a junction photodiode can be affected by two types of phenomena [2]: 1) transit-time effects, which arise from the delay between the absorption of the incident light and the separation of photo-excited electron-hole pairs, and 2) lumped-constant effects, which degrade the high-frequency performance because of inherent electrical characteristics such as the series resistance of the diode and its junction capacitance.

Considering the configurations presently adopted for

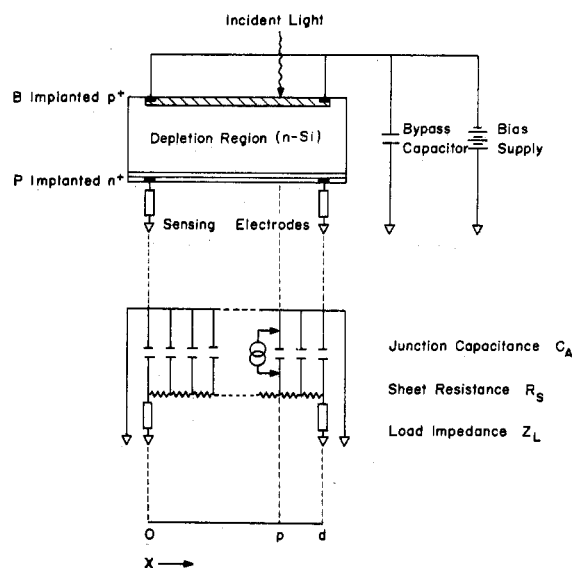


Fig. 1. Device structure, bias network, and equivalent circuit of position-sensitive ion-implanted lateral photodetector. Note that four electrodes are required to define a two-dimensional orthogonal coordinate system.

PSD's, transit-time effects can be assumed of little consequence, as will be shown later. The purpose of this paper, then, is to consider the impact of essential  $RC$ -type device parameters on the frequency response of lateral photodetectors. Specifically, we propose to demonstrate that a reasonably accurate description of relevant experimental observations can be developed on the basis of a simple model patterned upon the work of Kalbitzer and Melzer [3] for one-dimensional position-sensitive particle detectors.

## II. RESPONSE TIME: ANALYTICAL RESULTS

Fig. 1 illustrates the basic configuration of "fast" position-sensitive photodetectors developed at Raytheon. The design rests upon the utilization of an ion-implanted resistive layer on the nonilluminated face of the detector for the purpose of optimizing the collection and distribution of photo-excited carriers to four lateral sensing electrodes. The device is always operated in a fully depleted mode, which ensures that carrier-diffusion current components will be minimal, thus enhancing the effective quantum efficiency as well as the speed of response.

In order to discuss the pulse-response characteristics of these devices, we will make use of the equivalent circuit shown in Fig. 1, which is based on the assumption that a light-induced signal reaches the sensing electrodes by means of an analog transmission line involving portions of the continuously distributed junction capacitance and

Manuscript received October 1, 1973; revised April 15, 1974.  
The authors are with the Research Division, Raytheon Company, Waltham, Mass. 02154.

the homogeneous sheet resistance of the implanted backlayer. This simple model [3] is expected to hold for fully depleted devices with a thin highly doped resistive layer if the width of the depletion layer remains much smaller than the linear dimension of the detector in the p-n junction plane. Thus we may anticipate that the "lateral" response time, for a given light-impact position, reflects not only the distance from the relevant electrode, but also both the capacitance per unit area ( $C_A$ ) and the sheet resistance ( $R_S$ ) of the detector.

For the devices under consideration, the transit times of holes and electrons are short compared to the time constant associated with lateral transport. In other words, we may assume that upon instantaneous release of charges by an incident light pulse, the sequence of events leading to the discharge of a local infinitesimal capacitor (see Fig. 1) can be described in terms of an idealized initial condition. Furthermore, in describing the charge-distribution process, that is, the process in which charges that are initially concentrated at the position of light incidence get delivered to the sensing electrodes, we may exclude contributions to the series resistance stemming from any other source but the sheet resistance of the implanted backlayer. Thus the transient voltage distribution that governs the charge-collection process should obey a two-dimensional transmission-line equation such as

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = R_S C_A \frac{\partial V}{\partial t} \quad (1)$$

with appropriate boundary and initial conditions.

In order to obtain an analytical solution of relevance in elucidating the nature of the charge-collection process, and avoid a prohibitive amount of mathematical complexity, we may first consider the one-dimensional model of Fig. 1, that is, write

$$\frac{\partial^2 V}{\partial x^2} = R_S C_A \frac{\partial V}{\partial t} \quad (2)$$

instead of (1). With virtually grounded terminals ( $R_L \ll R_S$ ) located at stations  $x = 0$  and  $x = d$ , the boundary conditions are

$$V(0, p, t) = V(d, p, t) = 0 \quad (3)$$

for any impact position  $p$ . Regarding initial conditions, we take it that only a very small fraction of the junction area will be discharged by an incident laser light pulse, which suggests we write

$$V(x, p, 0) = (Q_0/C) \delta(x - p) \quad (4)$$

where  $Q_0$  represents the total electron charge released and  $C$  is the total junction capacitance. A formal solution of such a system of equations is available in the literature on heat flow [4] and indicates that the function

$$V(x, p, t) = \left(\frac{Q_0}{C}\right) \sum_{n=1}^{\infty} 2 \sin\left(\frac{n\pi x}{d}\right) \sin\left(\frac{n\pi p}{d}\right) \exp\left(-\frac{n^2 \pi^2 t}{RC}\right) \quad (5)$$

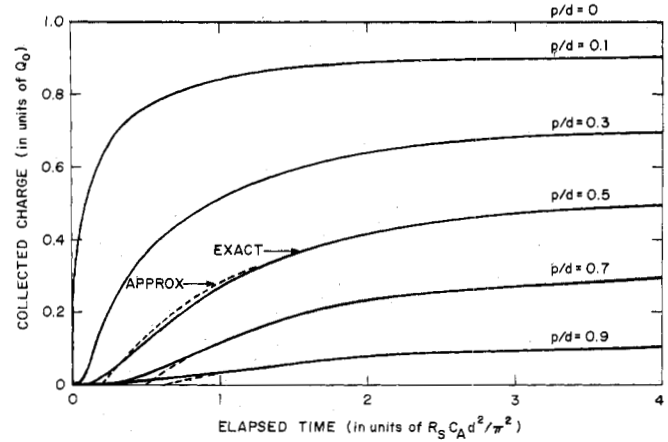


Fig. 2. Fractional amount of charge collected at the sensing electrode of one-dimensional position-sensitive detector, for various positions of initial instantaneous charge release. Time scale is expressed in units of fundamental time constant, which is used here to approximate exact response characteristics for impact positions at or beyond half-distance point to the opposite electrode. Note that, in this approximation, initial time delays range from about 0.2 to 0.6 times fundamental time constant.

satisfies all the conditions of the problem if it is understood that

$$RC = R_S C_A d^2. \quad (6)$$

On this basis, and following Kalbitzer and Melzer [3], we can determine how locally released charges are actually delivered to PSD terminals. Specifically, we have

$$Q(0, p, t) = Q_0 \sum_{n=1}^{\infty} \left(\frac{2}{n\pi}\right) \sin\left(\frac{n\pi p}{d}\right) \left[1 - \exp\left(-\frac{n^2 \pi^2 t}{RC}\right)\right] \quad (7)$$

which describes the kinetics of the charge-collection process within the framework of the idealized model.

Fig. 2 illustrates an evaluation of (7) in terms of the fraction of the total charge collected as a function of the reduced time  $t/(RC/\pi^2)$ , for positions of incidence ranging from  $p = 0$  to  $p = d$ . In accordance with the model, there is no delay in collecting charges released at  $x = 0$ ; for all other impact positions, however, the charge buildup experienced at the sensing electrode exhibits rapidly increasing response times. Note that beyond the half-distance point to the opposite electrode, the risetimes (10 to 90 percent) appear to stabilize at a level roughly equal to  $2.2 RC/\pi^2$ , which suggests to approximate the limiting pulse response of lateral photodetectors by means of a fundamental single-pole  $RC$ -filter time constant,

$$\tau_c = R_S C_A d^2 / \pi^2 \quad (8)$$

in conjunction with position-dependent initial delay times of up to  $0.6\tau_c$  (see Fig. 2).

In this connection, reference must be made to the calculations by Connors [5], which illustrate how lateral photodetectors respond to a step input of radiation, on assuming a one-dimensional geometry. It can be seen that the short-circuit current risetime is dependent on the position of light incidence and reaches values of about

POSITION-SENSITIVE PHOTODETECTOR: UNIT 542.05

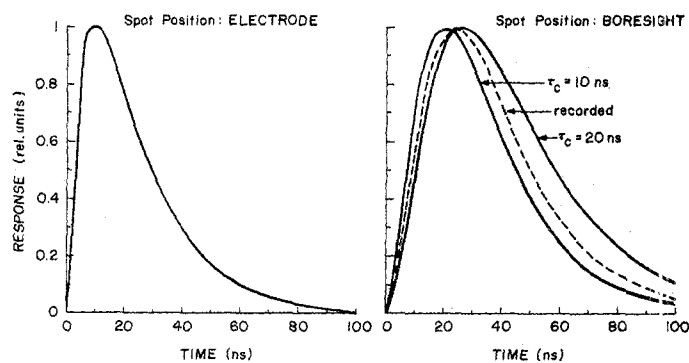


Fig. 3. Nd:YAG laser-pulse response characteristics of ion-implanted position-sensitive photodetector described in Table I. Curves labeled  $\tau_c = 20$  ns and  $\tau_c = 10$  ns are calculated waveforms obtained through convolution of "on-electrode" and "impulse" responses for fundamental time constants of 10 and 20 ns, respectively. Note that all waveforms are normalized and, thus, do not reflect the falloff in peak amplitude associated with an off-electrode incidence.

$0.25R_s C_A d^2$  (in our notations) as the distance from the sensing electrode exceeds  $d/2$ . Thus it would appear that, based upon Connor's results and disregarding initial delay times, the limiting time constant of a lateral photodetector should be on the order of  $0.1R_s C_A d^2$ , or essentially as given by (8). Furthermore, it should be mentioned that, according to Lucovsky and Emmons [6], the frequency at which the effect of the sheet resistance of the diffused region of a conventional p-n junction photodiode becomes significant is given by

$$1/(2\pi f_c) = N(\rho/w) C_A b^2 \quad (9)$$

where  $f_c$  refers to the 3-dB cutoff frequency of the short-circuited diode,  $\rho/w$  is the analog sheet resistance, and  $b$  the distance between contact and farthest point on the diode face. Since the parameter  $N$  represents a number dependent upon the diode geometry we may conclude that an *Ansatz* such as  $\tau_c = N(R_s C_A d^2)$  may provide a valid approach to describing the frequency response of two-dimensional lateral photodetectors, the parameter  $N$  somehow reflecting the configuration of the electrode arrangement.

### III. RESPONSE TIME: EMPIRICAL RESULTS

Fig. 3 refers to an ion-implanted two-dimensional PSD sensor element made at our laboratory, and illustrates its response to Q-switched Nd:YAG laser pulses. The focused laser beam was first directed to a spot over an electrode, then moved to a point approximately at the center of the element, thus generating output signals as shown on the left-hand and right-hand sides of Fig. 3, respectively. These tests indicate that the input pulse shape is preserved, but there is evidence of pulse stretching and of differential time delays between pulse peaks as the spot shifts from an "electrode" to a "boresight" position within the detector active area. An assessment in

TABLE I

|                       |                                                        |
|-----------------------|--------------------------------------------------------|
| Bulk material:        | n-type silicon (4000 $\Omega$ -cm)                     |
| Wafer thickness:      | 10 mil (250 $\mu$ m)                                   |
| Implanted junction:   | 35-keV boron ( $5 \times 10^{15}$ cm $^{-2}$ )         |
| Back layer:           | 150-keV phosphorus ( $1.5 \times 10^{13}$ cm $^{-2}$ ) |
| Electrode separation: | 6 mm, each pair                                        |
| Sheet resistance:     | 9 k $\Omega$ per square                                |
| Junction capacitance: | 0.42 pF/mm $^2$ (a)                                    |
| Bias voltage:         | 100 V                                                  |
| Load impedance:       | 50 $\Omega$ , each electrode                           |

Note: Device characteristics of a Raytheon/Research lateral photodetector (unit 542-05) designed for Nd:YAG laser-pulse position sensing.

<sup>a</sup> Assumes a fully depleted operational regime.

terms of the lateral time constant of this device requires information on design parameters, which is summarized in Table I.

First, we must establish that drift transit times do not seriously affect the speed of response of this fully depleted unit when responding to 20-ns wide incident laser pulses. On assuming that the electron transit time  $\tau_n$  represents the limiting factor in this regard [7], it is readily seen that the transit times can be expected not to exceed 5 ns, which implies that the relevant bandwidth ( $1/\tau_n$ ) is at least 200-MHz wide. Consequently, even if some bandwidth reduction should occur because of the finite load resistance, it follows that on "illuminating" an electrode the detector output assumes a shape close to the driving pulse waveform.

Let  $V_0(t)$  represent the input waveform and  $U(t)$  the impulse response of the detector; specifically, we take it that the function  $U(t)$  characterizes the detector response to a delta-shaped excitation pulse incident at or near the boresight position. By virtue of the convolution principle, the output waveform  $V_1(t)$  then can be expressed as

$$V_1(t) = \int_0^t V_0(t') U(t - t') dt' \quad (10)$$

which allows us to infer the shape of the output signal generated by an input  $V_0$  if the impulse-response function is available. In view of the results of Section II, but disregarding initial time-delay considerations, it should be a reasonable approximation to write  $U(t) = \exp(-t/\tau_c)$ , where  $\tau_c$  refers to the fundamental time constant of the lateral photoeffect. Furthermore, we find that an excellent description of the "input" as illustrated on the left-hand side of Fig. 3 is provided for by

$$V_0(t) = \exp(-t/a) - \exp(-t/b) \quad (11)$$

with  $a = 18$  ns and  $b = 6$  ns. On this basis, the output waveform for laser light pulses impacting in the vicinity of the center location of the detector active area can be

obtained through integration of (10), which yields

$$V_1(t) = \frac{\exp(-t/\tau_c) - \exp(-t/a)}{1/a - 1/\tau_c} - \frac{\exp(-t/\tau_c) - \exp(-t/b)}{1/b - 1/\tau_c} \quad (12)$$

where the time constant  $\tau_c$  is an adjustable parameter. Fig. 3 (right-hand side) shows two calculated waveforms corresponding to  $\tau_c = 10$  ns and  $\tau_c = 20$  ns, thus illustrating how lateral photoeffects may shift the pulse-peak position and stretch the width of the undistorted pulse. The actual pulse shape as recorded on a Type 454 oscilloscope is also plotted in Fig. 3, the comparison pointing to a fundamental response time  $\tau_c$  of, say, 14 ns. Returning now to the *Ansatz* suggested in connection with (9), and making use of the values listed in Table I for sheet resistance, capacitance per unit area, and electrode separation, we conclude that for the type of device under consideration the parameter  $N$  should be close to 1/10. In other words, from the point of view of device design criteria it would appear that (8) provides a simple and reasonably accurate description of the frequency-response characteristics of two-dimensional PSD's.

#### IV. DISCUSSION AND APPLICATION

In summary, then, we may express the fundamental bandwidth of a two-dimensional position-sensitive photodetector in a manner such as

$$f_c = \frac{\pi/2}{R_s C_A d^2} \quad (13)$$

which clearly identifies the critical design parameters in problems involving the detection and localization of incident very short duration light signals. Further comments are in order:

1) Lowering the sheet resistance of the nondepleted backlayer through ion-implantation or any other technique provides a straightforward means of minimizing rise-time effects. It should be kept in mind, however, that high sheet resistances are desirable for limiting the Johnson noise in the detector/preamplifier output and, hence, that an optimum value of  $R_s$  must be chosen on the basis of tradeoffs that are beyond the scope of the present paper [8].

2) A reduction of the capacitance  $C_A$  is accomplished by achieving wider depletion regions, which implies using thicker wafers of high-resistivity silicon compatible with highest practical bias voltages. Here again, there are constraints imposed by transit-time considerations, and an optimum solution must take the resulting degradation into account together with the potential benefit of higher quantum efficiencies associated with wider depletion layers.

3) The spacing of the signal electrodes strongly affects the bandwidth, and care must be exercised if the dependence of pulse shape on image position is to be minimized.

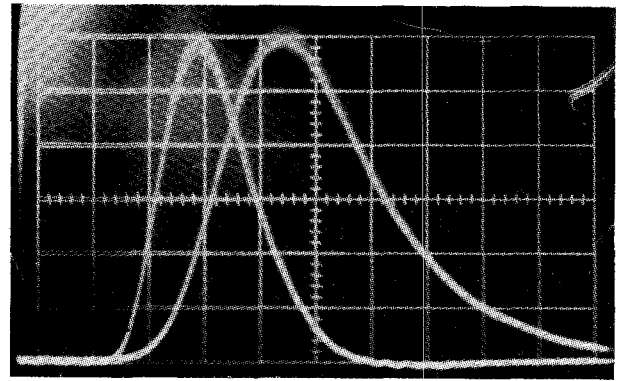


Fig. 4. Pulse-response characteristics of typical position-sensitive photodetector for nominally 20-ns wide GaAs laser inputs. Note how FWHM increases from about 18 to almost 32 ns as position of incidence shifts from over the electrode to center location. (Horizontal scale: 10 ns/division; vertical scale: arbitrary.)

In this regard, we may emphasize that a wider separation of the electrodes allows for increased optical image formats and, thus, wider fields of view but the position-sensing function will be degraded whenever the cutoff frequency  $f_c$  becomes inherently too low for detecting and processing narrow laser-light pulses.

To examine this situation, we now consider a large-format circular PSD device having a capacitance of approximately 100 pF. From signal-to-noise ratio considerations it follows that for 20-ns wide incident light pulses the detection bandwidth should be at least equal to 10 MHz, which means that the backlayer sheet resistance should not exceed 1 k $\Omega/\square$ . The amount of pulse stretching experienced under these conditions is portrayed in Fig. 4, the narrow trace [full width at half maximum (FWHM) = 18 ns] reflecting the input waveform from a GaAs laser diode as seen at the signal electrode, whereas the wide trace (FWHM = 32 ns) shows the "scaled-up" detector response for a boresight-type illumination. The reader may also note the effect of the initial delay time, which is on the order of 5 ns, thus pointing (see Fig. 2) to a fundamental time constant of about 20 ns, in essential agreement with response characteristics expected from a device of this nature.

Further results of interest are shown in Figs. 5 and 6, which illustrate the dependence of pulse response on spot position for two representative device structures. Specifically, Fig. 5 refers to a small-format type PSD sensor with

$$d = 5 \text{ mm}, \quad C_A = 0.42 \text{ pF/mm}^2, \quad R_s = 5 \text{ k}\Omega/\square$$

and displays the response to Nd:YAG laser pulses for three positions of light incidence: over the signal electrode, at the center location, and near the far edge. Clearly, there is no evidence of degradation in the pulse response, as may have been anticipated considering that the fundamental bandwidth of this device should be close to 30 MHz. Fig. 6, on the contrary, shows some very substantial stretching of the input waveform as the spot moves from the reference electrode to the opposite edge of a sensor

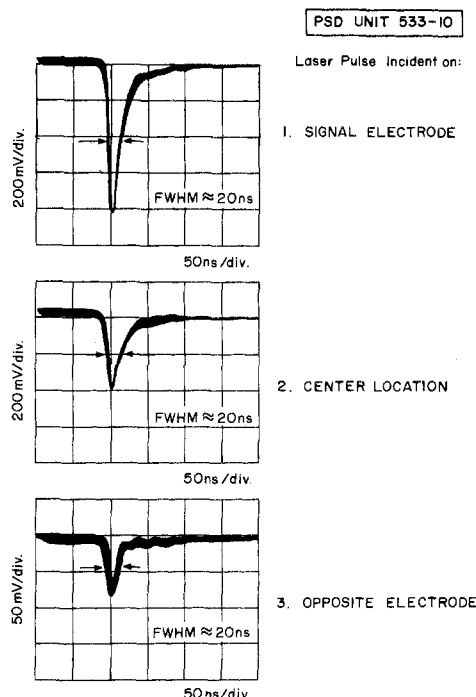


Fig. 5. Nd:YAG laser-pulse response characteristics of small-format position-sensitive photodetector, as measured across 50- $\Omega$  terminal load resistance. Note the presence of "tail" oscillations, which are believed related to insertion of recording circuit.

with parameters as follows:

$$d = 20 \text{ mm}, \quad C_A = 0.42 \text{ pF/mm}^2, \quad R_S = 2 \text{ k}\Omega/\square.$$

Since the total junction capacitance is about 130 pF, the bandwidth of this device should not extend beyond 5 MHz thus inducing considerable distortion of 18-ns wide GaAs laser pulses. We note that relatively little additional stretching occurs as the spot moves from a central location to the opposite electrode (FWHM = 56 ns versus 50 ns) thus confirming the pattern disclosed in Fig. 2. Furthermore, the initial delay of almost 20 ns in the onset of the response for a far-edge illumination turns out to be precisely  $0.6\tau_c$ , with  $\tau_c = 32$  ns, again in accordance with the approximation of Fig. 2. Finally, it is worth emphasizing how pulse stretching affects the relative amplitude of the oscilloscope traces reproduced in Figs. 5 and 6,<sup>1</sup> thus affecting the position-sensing function of lateral photodetectors operating in a ratiometric signal-processing mode.

## V. CONCLUSION

In designing position-sensitive photodetectors for laser-systems applications, it is essential to observe that frequency response characteristics are controlling factors not only from the point of view of the ultimate pulse-resolution performance, but also in terms of the lateral falloff and, thus, the effective operation of these devices. Specifically, in dealing with narrow laser pulses, and on

<sup>1</sup> The center-to-electrode ratio drops from a nominal value of 1/4 to less than 1/10.

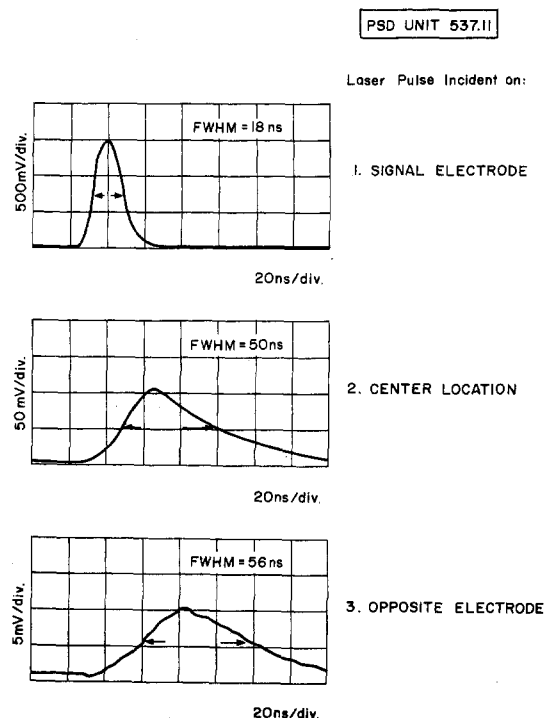


Fig. 6. GaAs laser-pulse response characteristics of large-format position-sensitive photodetector, as seen at output terminals of broad-band low-input impedance preamplifier.

assuming fixed electrode spacings, faster response times lead to lower values of the falloff parameter [1], hence improving the apparent sensor responsivity at boresight incidence. It has been the object of this paper to examine how device-design parameters determine the pulse-response characteristics of PSD's, and to establish that reasonably accurate predictions can be made in the frame of simple analytical considerations.

1) We have demonstrated that the model of Kalbitzer and Melzer [3] provides an adequate basis for describing the charge-collection kinetics in fully depleted lateral photodetectors optimized for laser-pulse position sensing. It appears that the limiting pulse-response behavior can be approximated by means of a fundamental  $RC$ -filter time constant  $\tau_c$  together with position-dependent initial delay times of up to  $0.6\tau_c$ .

2) This approach has been utilized to derive through convolution the shape of the detector-output waveform for laser light pulses impacting in the vicinity of the center location. A comparison with recorded pulse-response traces then suggests that the response time  $\tau_c$  of two-dimensional PSD's should be close to a theoretical value of  $R_S C_A d^2 / \pi^2$ , thus implying that the limiting bandwidth of practical units with conventional electrodes should obey an equation such as (13).

3) This result identifies the critical design parameters and provides a quantitative base for performing the required tradeoffs in terms of thermal noise considerations as well as carrier transit times. Illustrations of the dependence of pulse response on spot position for representative ion-implanted lateral photodetectors developed at Ray-

theon appear to substantiate some inherent implications of the analysis in the context of evaluating position-sensitive detectors for applications involving very short laser pulses.

### ACKNOWLEDGMENT

The authors wish to thank G. DeMars for fabricating the detectors and S. Thomas for supplying some of the oscilloscope traces used in assessing their performance; thanks are also due to K. Kinnard for stimulating discussions on systems related aspects of position sensing via lateral photoeffects in silicon.

### REFERENCES

- [1] T. G. Carr, T. C. Richmond, and R. G. Wagner, "Position-sensitive Schottky barrier photodiodes: time-dependent signals

- and background saturation effects," *IEEE Trans. Electron Devices*, vol. ED-17, pp. 507-513, July 1970.
- [2] H. Melchior, M. Fisher, and F. Arams, "Photodetectors for optical communication systems," *Proc. IEEE*, vol. 58, pp. 1466-1486, Oct. 1970.
- [3] S. Kalbitzer and W. Melzer, "On the charge dividing mechanism in position-sensitive detectors," *Nucl. Instrum. Methods*, vol. 56, pp. 301-304, Nov. 1967.
- [4] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*. London, England: Oxford Univ. Press, 1959, ch. 3.
- [5] W. P. Connors, "Lateral photodetector operating in the fully reverse biased mode," *IEEE Trans. Electron Devices*, vol. ED-18, pp. 591-596, Aug. 1971.
- [6] G. Lucovsky and R. Emmons, "Lateral effects in high-speed photodiodes," *IEEE Trans. Electron Devices*, vol. ED-12, pp. 5-12, Jan. 1965.
- [7] M. V. Schneider, "Schottky barrier photodiodes with anti-reflection coating," *Bell System Tech. J.*, vol. 45, pp. 1611-1638, Nov. 1966.
- [8] E. Laegsgaard and F. Martin, "Position-sensitive semiconductor particle detectors fabricated by ion implantation," *IEEE Trans. Nucl. Sci.*, vol. NS-15, pp. 239-245, June 1968.

## Correspondence

### Measurements on Light Emission and Current Distribution in Avalanche Epitaxial Diodes

A. BAR-LEV AND H. AHARONI

**Abstract**—Silicon diodes built in epitaxial layers of varying parameters were found to exhibit two distinct modes of light-emission dependence on avalanche current prior to secondary breakdown. The modes depend on whether the depletion layer extends to the substrate interface prior to avalanche or not. Current distribution is inferred from the light curve and a possible mechanism leading to the secondary breakdown is suggested.

The purpose of this letter is to describe the light-emission behavior of avalanche silicon epitaxial diodes up to the onset of secondary breakdown (SB) [1] and to correlate it with current-density variations across the diode and with the epitaxial layer resistivity and thickness parameters.

The diodes were made by the regular bipolar planar process. An epitaxial n-type layer (phosphorous doped), of the desired thickness and resistivity, was deposited on 0.01  $\Omega\cdot\text{cm}$  (Sb doped) substrate and then subjected to a two-step boron diffusion process. The junction area and depth were  $3 \times 10^{-3} \text{ cm}^2$  and  $2.85 \pm 0.15 \mu\text{m}$ , respectively. Two series of diodes were made. The first had epitaxial layers with thicknesses varying between 6.3 to  $17.2 \mu\text{m}$  while their resistivity was kept constant at  $0.485 \pm 0.025 \Omega\cdot\text{cm}$ . The second consisted of diodes built on layers of constant thickness of  $12.2 \pm 1 \mu\text{m}$  and varying resistivities of 0.1 to  $10.5 \Omega\cdot\text{cm}$ . Each series contained several hundred diodes and only those whose avalanche voltage approximately agreed with the value predicted from resistivity measurements were included in the tests. Thus it can be assumed that no major crystal faults occur within their area. The diodes can be divided into two groups which do not parallel the two series. The first is characterized by a depletion layer that extends to the epitaxial layer-substrate interface before avalanche starts. Those

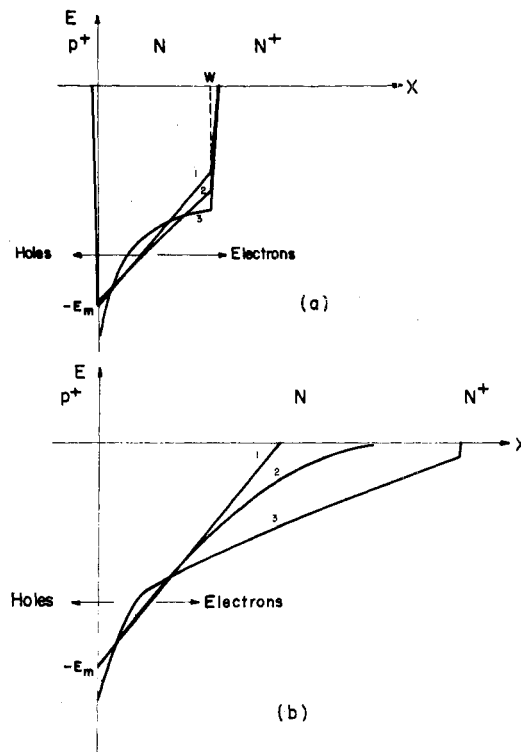


Fig. 1. Field distribution in depletion layer. Curve 1: negligible free carrier densities. Curve 2: medium avalanche current density. Curve 3: high avalanche current density. (a) Punch-through case. (b) Non-punch through case.

will be called the punched-through group. Their depletion layer field pattern can be obtained from Poisson's equation using step junction theory and is shown by curve 1 of Fig. 1(a) for the pre-avalanche case. The second group consists of nonpunch-through diodes whose preavalanche field pattern is shown by curve 1 of Fig. 1(b).

Manuscript received November 18, 1973; revised March 21, 1974.  
A. Bar-Lev is with the Technion—Israel Institute of Technology, Haifa, Israel.  
H. Aharoni is with the University of the Negev, Beer-Sheva, Israel.